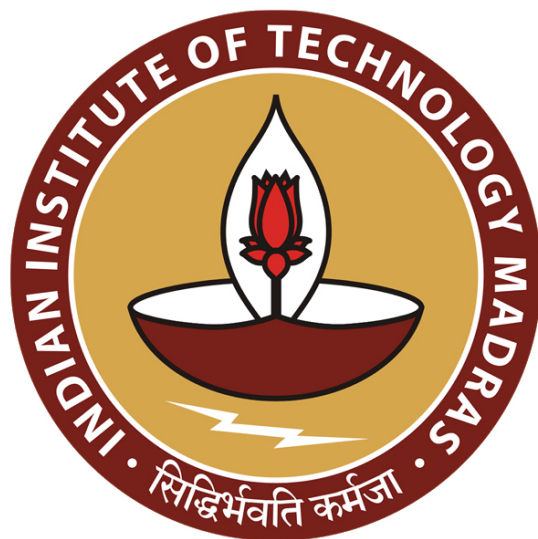




IIT Madras
ONLINE DEGREE

Database Management Systems
Professor Partha Pratim Das
Department of Computer Science and Engineering
Indian Institute of Technology, Kharagpur
Storage and File Structure/I: Physical Storage

Welcome to Module 39 of Database Management Systems course in IIT Madras online BSc program.

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Module Recap

- Introduced Non-linear Data Structures - graph, tree, hash table
- Studied Binary Search Tree as an adaptation of binary search
- Compared Linear and Non-Linear Data Structures

Database Management Systems

Module Objectives

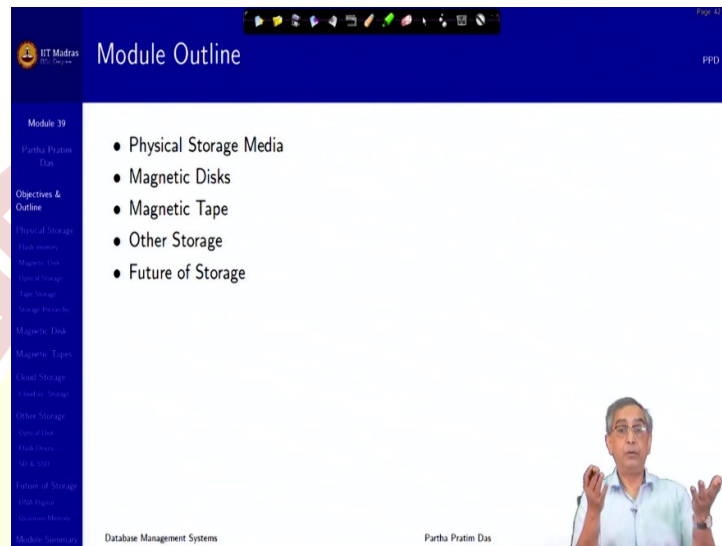
- Introduce various Physical Storage Media for high volume, fast, reliable and inexpensive options for data storage for databases
- To understand the options of Tertiary Storage for high volume, inexpensive backup options

Database Management Systems

We have in the last three modules discussed about various algorithms and data structures linear as well as nonlinear for management of data primarily in-memory. Now, what we are going to get into is the management of storage, which is directly required for our database system, and the primary factor in this is along with the in-memory data structure because you need them to be able to do the operations while you are updating a database or during a query

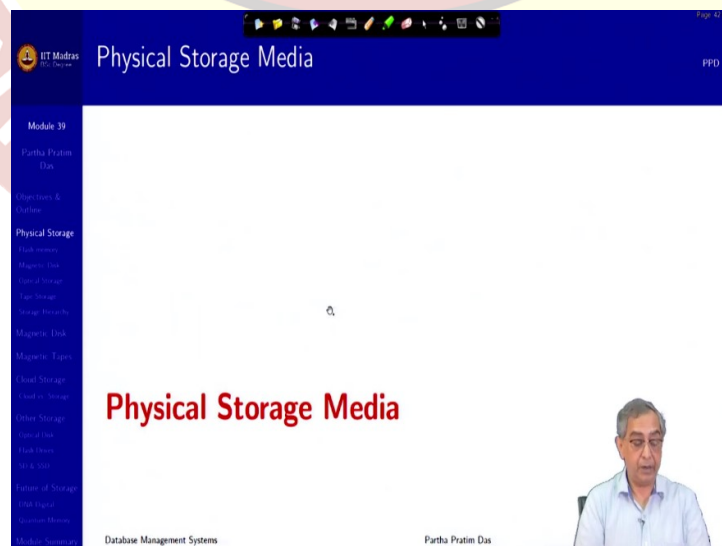
and so, on along with that, what you significantly need is a large volume of persistent data storage, which does not go away if your program has finished or not. And you also you need archivals that data becomes larger and larger and you need to keep it for the future archive. So, we are going to start discussing about all those.

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So, what they give here is a glimpse of the physical, different forms of physical storage media that are there. Some we will discuss in little bit depth like magnetic disk, and some we will just hear because I want you to know the entire gamut of things and also in the future in the next couple of years what do you expect to be coming on our way.

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Classification of Physical Storage Media

- Speed with which data can be accessed
- Cost per unit of data
- Reliability
 - data loss on power failure or system crash
 - physical failure of the storage device
- Can differentiate storage into:
 - **volatile storage**: loses contents when power is switched off
 - **non-volatile storage**:
 - ▷ Contents persist even when power is switched off
 - ▷ Includes secondary and tertiary storage, as well as battery-backed up main-memory

So, physical storage media, what are the primary factors that we have to consider? Obviously, speed is the first factor, how quickly it can be accessed because database is large. If we take more time, the whole operation will become slow. Naturally the second is, cost per unit of data per byte of data or per megabyte of data or per gigabyte of data, that cost has to be as low as possible, because we are going to have very, very large data storage.

Then we are looking for persistency, so the reliability is very important. The power can fail, the system might crash, there can be physical failure in the storage device, will the data be protected against all these. So, for this whole discussion, that there are two parts that we will have to look at. One is the volatile storage, which is typically the computing memory, what we were saying as in-memory and non-volatile storage, which will stay.

(Refer Slide Time: 03:03)

Physical Storage Media

- **Cache**
 - fastest and most costly form of storage
 - volatile
 - managed by the computer system hardware
- **Main memory**
 - fast access (10's to 100's of nanoseconds (ns); $1 \text{ ns} = 10^{-9} \text{ seconds}$)
 - generally too small (or too expensive) to store the entire database
 - ▷ capacities of up to a few Gigabytes widely used currently
 - ▷ Capacities have gone up and per-byte costs have decreased steadily and rapidly (roughly factor of 2 every 2 to 3 years)
- **Volatile**
 - contents of main memory are usually lost if a power failure or system crash occurs

So, talking about the volatile storage, you have the cache as a fastest memory on your system. Besides your registers, besides your CPU registers, which are really few. So, it is a very, very fast and most costly form of storage, so you can have only a small amount of it maybe that could be hierarchy also.

Then you have the main memory, which is also volatile. It has fast access, but not as fast as a cache. Typically, tens to hundreds of nanoseconds, and usually, but it is usually too small or make it too expensive to store the entire database in the memory, so it has to manage by taking only a part of the database do the manipulation and then again put it back to the persistent data, and volatility is the main factor.

(Refer Slide Time: 03:59)

The screenshot shows a presentation slide from IIT Madras titled "Physical Storage Media (2): Flash memory". The slide is part of "Module 39" and lists the following points:

- Data survives power failure
- Data can be written at a location only once, but location can be erased and written to again
 - Can support only a limited number (10K – 1M) of write/erase cycles
 - Erasing of memory has to be done to an entire bank of memory
- Reads are roughly as fast as main memory
- But writes are slow (few microseconds), erase is slower
- Widely used in embedded devices such as digital cameras, phones, and USB keys

The slide also includes a small video inset of a man speaking in the bottom right corner. The footer of the slide reads "Database Management Systems" and "Partho Pratim Das".

Now, you have something intermediate coming up these days have come up these days which are flash memories. So, flash memories are those where, which are like the main memory, but they survive on power failure. Typically, the data can be written at a location only once, but it can be erased and written to again. It is a little different kind of technologies. It is not that you can read write, read write, read write, you cannot do that. You can write it once, and then you can do an operation called erase, and that need, that cannot be done at an isolated location that has to be done for the entire bank of memory.

And you can then again write to it, but also you cannot do this in finite number of times possibly a flash memory will survive 10,000 erase cycles or maybe a million erase cycles, but not indefinite. But this is a good alternate in many places as an extensional larger main memory, because that is one which survives the power failure. It is, so it is in performance it

is like the main memory, in cost it is not that high, but in terms of persistency, it has some persistency against power failure.

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Physical Storage Media (3): Magnetic Disk

- Data is stored on spinning disk, and read/written magnetically
- Primary medium for the long-term storage of data ✓
 - typically stores entire database
- Data must be moved from disk to main memory for access, and written back for storage - much slower access than main memory
- **direct-access** ✓✓✓
 - possible to read data on disk in any order, unlike magnetic tape
- Capacities range up to roughly 16-32 TB
 - Much larger capacity and much lower cost/byte than main memory/flash memory
 - Growing constantly and rapidly with technology improvements (factor of 2 to 3 every 2 years)
- Survives power failures and system crashes
 - disk failure can destroy data, but is rare

Now, what is the mainstay for storage of persistent data is magnetic disks. These are basically, I mean, we will get to the details, but it is their collection of I am sorry. They are a collection of spinning disks where read/write is done magnetically and the magnetic memory remembers that. It is used for long term storage and but it cannot be directly operated on. The only thing you can do is go and write there and read back, access, that is it. Much slower than main memory, so you have to bring that data from the disk to the main memory and then operate on it, put it back.

The big advantage of magnetic disk is direct access, which means, in a very large database in a magnetic disk, you can go to any part of the data and you can read that. You do not have to, there is no seriality, serialization in that, so you do not have to go to pass through the earlier data like, so direct access is similar in concept to arrays, which we say is a random access, it is not as fine grained as random access that you can randomly go to a byte, but you can go to different sectors, pages and so on, but you can directly go there without going through anything else, which is in contrast to what you do in a linked list kind of structure, which we will call it sequential access, where unless you read the previous data items, you cannot read the later on.

So, this is, this property makes it persistence and this property makes it really attractive, as a store for the main store for the databases. Capacities could be larger, it runs typically 16 to 32 terabyte and every two to three years capacities typically double. It survives power failure, system crash, but if the disk itself fails then it will destroy the data, but that is very, very rare, we will talk about that.

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Physical Storage Media (4): Optical Storage

- non-volatile, data is read optically from a spinning disk using a laser
- CD-ROM (640 MB) and DVD (4.7 to 17 GB) most popular forms
- Blu-ray disks: 27 GB to 54 GB
- Write-one, read-many (WORM) optical disks used for archival storage (CD-R, DVD-R, DVD+R)
- Multiple write versions also available (CD-RW, DVD-RW, DVD+RW, and DVD-RAM)
- Reads and writes are slower than with magnetic disk
- **Juke-box** systems, with large numbers of removable disks, a few drives, and a mechanism for automatic loading/unloading of disks available for storing large volumes of data

Database Management Systems | Partha Pratim Das

There are certain forms of optical storage, but most of them as you have been using the CD ROMs, the DVDs, the Blue-ray and so on are primarily write-only device. You just write once and read many times, so they are good for archival system, but not as a live database storage system.

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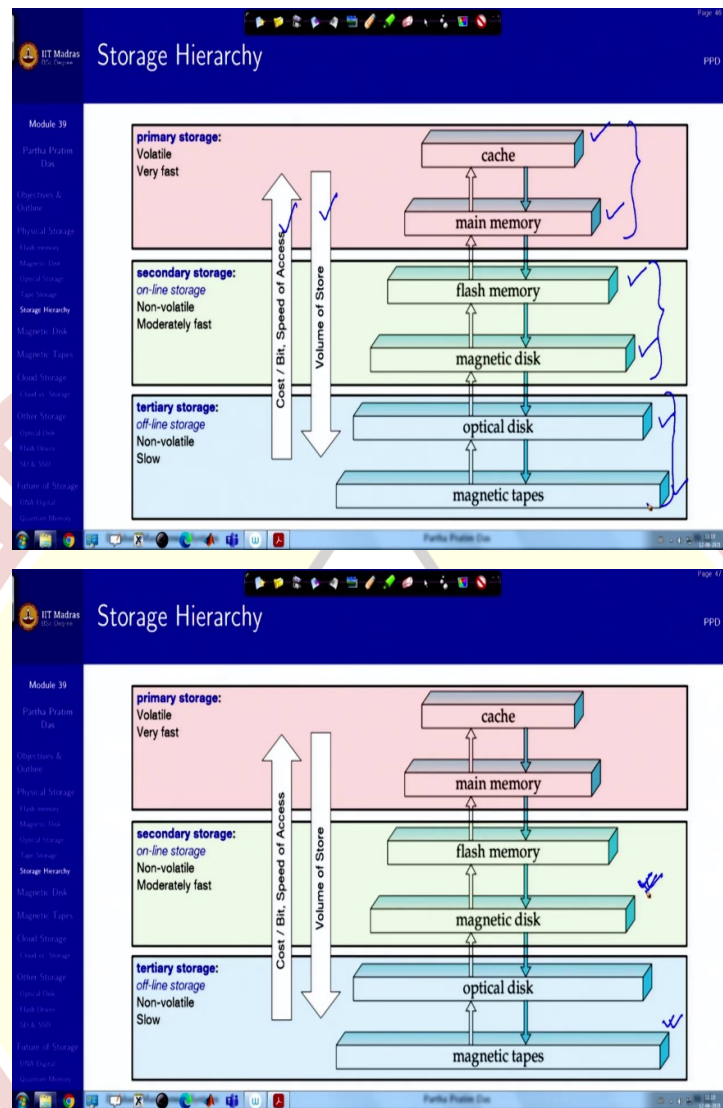
Physical Storage Media (5): Tape Storage

- non-volatile, used primarily for backup (to recover from disk failure), and for archival data
- **sequential-access**
 - much slower than disk ✓
- very high capacity (40 to 300 TB tapes available) ✓
- tape can be removed from drive storage costs much cheaper than disk, but drives are expensive
- Tape jukeboxes available for storing massive amounts of data
 - hundreds of **terabytes (TB)** ($1 \text{ TB} = 10^{12} \text{ bytes}$) to even multiple **petabytes (PB)** ($1 \text{ PB} = 10^{15} \text{ bytes}$)

Then you have tape storage. Tapes are, again, they are non-volatile, they are as I mean, you have seen the audio tapes. I hope you have, because they are rarely seen these days. So, these are basically long tapes where you keep on writing the data. The disadvantage, main disadvantage of tape, as to why you cannot use it on a live database system is the fact that it has sequential access like it is a tape.

So, if you have written the data in the 100th position, you will have to skip the first 99 data items to read the 100th. If it is at 1 millionth position you will have to skip 1 million minus one number of data items. So, that makes it much slower because you do not have a direct access, but on the other side it has the advantage of having much larger capacity, it has much lower cost per unit compared to the disks, and there are jukeboxes of tape available which are of hundreds of terabytes or even multiples of petabytes. So, for again, archival storage tape often is a very, very good idea.

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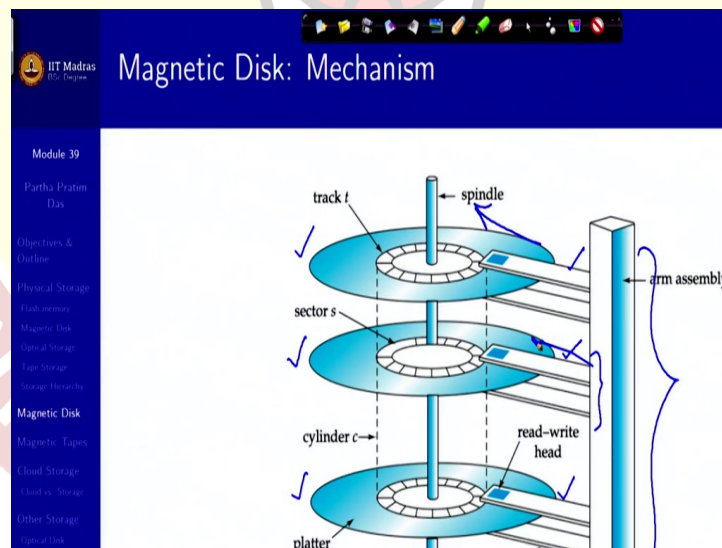


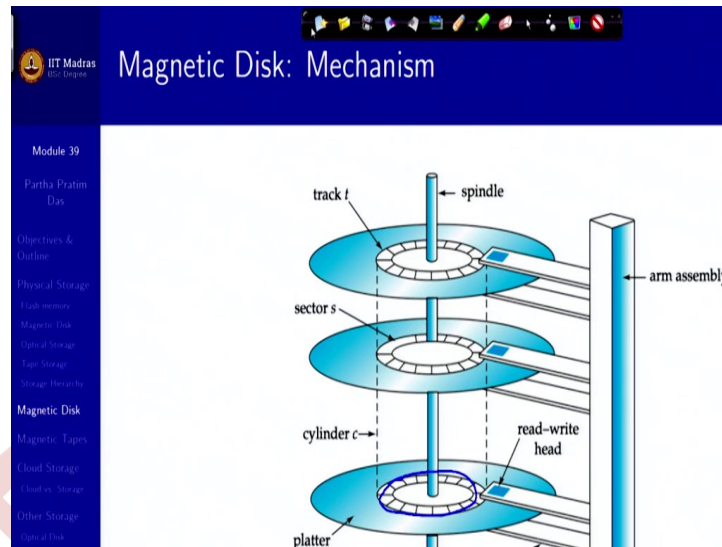
So, if I look into the whole gamut that we have just quickly run through, we just go through this on the top excluding the registers which are for computing. On the top we have cash, the fastest main memory followed by flash, magnetic disk, optical disk, if it is there as a, and the magnetic tapes.

We call this as primary memory, which means they are very fast, volatile, small, expensive, you can see those two these two, but is very needed for your regular computation. This is called secondary because this is where you keep your live database data, which is intermediately expensive can be very, can be quite large and much less expensive than the cache or main memory.

And then you have this tertiary storage, which are for offline storage, as I say for archival purposes, which are optical disks and magnetic tapes and so on. So, primarily, we are going to focus on this part and this part, primarily on this because that is what gives us the main stay for management of storage and performance of a database.

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So, onto magnetic disks. We are going to please study this part very, very carefully study the book very well on this, because this is going to be used in a whole spectrum of physical design aspects. So, this is a basic physical architecture of a typical magnetic disk, you can see there are disks. If you have been fortunate to see some earlier long playing records or things like that, those are like this. They are both sided, this is the, these are the read write heads. So, the heads as you can see are a pair because they if I have the discs like this, the head goes in like this, so there is one on the top, which is and one on the so both surfaces can be read at the same time when it is at the same position.

Now, so you have a number of discs on the spindle, which keeps on rotating at a very high speed, so that you can do so many, so many arms assembly of read writes can work in parallel. Now, since this is, this whole thing is rotating in together, when you position these disk heads, you position them at the same point. It is not that this will be more inside and this will be less inside, they are coupled, they go together.

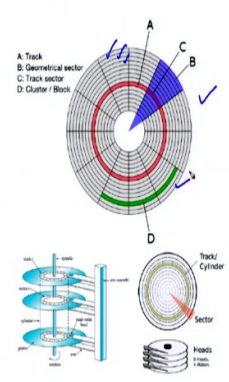
So, what happens is, when at any position point they are as if virtually, if you think of they are looking at a virtual cylinder, where you have circles in every disk two sides of every disk, and if you connect them, virtually, you see is a cylinder. So, you can think of that as one unit given one position of the head, if we change the position, you have a different cylinder. So, that is the basic stuff. Let us get into some more details.

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Magnetic Disk (2): Mechanism

- Read-write head
 - Positioned very close to the platter surface
 - Reads or writes magnetically encoded information
- Surface of platter divided into circular tracks
 - Over 50K-100K tracks per platter on typical hard disks
- Each track is divided into sectors
 - A sector is the smallest unit of data read or written
 - Sector size typically 512 bytes
 - Sectors / track: 500 to 1k (inner) to 1k to 2k (outer)
- To read/write a sector
 - disk arm swings to position head on right track
 - platter spins: Read/Write as sector passes under head
- Head-disk assemblies
 - multiple disk platters on a single spindle (1 to 5 usually)
 - one head per platter, mounted on a common arm.
- Cylinder i consists of i^{th} track of all the platters

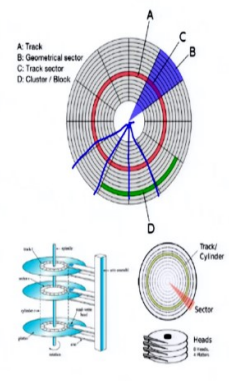


The diagram illustrates the mechanical components of a magnetic disk. It shows a top-down view of a disk platter with concentric circular tracks labeled A, B, C, and D. A specific track (A) is divided into sectors. A legend identifies: A: Track, B: Geometrical sector, C: Track sector, and D: Cluster / Block. Below this, a side view shows a disk arm with a read/write head positioned over the platter surface. Another view shows a stack of platters on a common spindle, with a head-disk assembly (HDA) moving along the spindle to access different cylinders.

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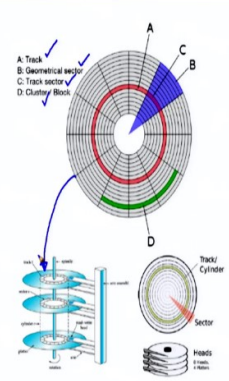


This slide is identical to the previous one, showing the same diagrams and text describing the mechanism of a magnetic disk, including the layout of tracks and sectors, the movement of the read/write head, and the structure of the head-disk assembly.

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This slide is also identical to the previous ones, providing the same detailed explanation and diagrams of magnetic disk mechanics, from the physical layout of the platter to the operation of the head-disk assembly.

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Magnetic Disk (2): Mechanism

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So, what you have in in a disk is, if you if you look at the surface of one disk, you have tracks, these are called tracks. You use the word track very much in context of music players and all that, but tracks actually came from that, that you have concentric circles, which basically can keep data. And as you can see that the outer tracks are naturally longer than the inner tracks, that is the basic geometry.

Now, given this track, you from the center, you think of radial lines. So, what will happen? Every track will get segmented and those segments are called. So, you have tracks and those segments are called sectors. So, you can have multiple sectors in a track. Now, naturally as you do this, the inner track has a shorter length, outer track has a longer length. So, what will happen is, number of sectors per track will differ.

So, what you do is, keep every sector size fixed, make it 512 bytes, let us say. Just for the sake of it there could be different, but sectors per track will vary maybe it is 500 to 1,000 in a inner track, whereas, it could be 2,000 to 1,000 to 2,000 for a outer track because the track is longer. So, those are those are your tracks.

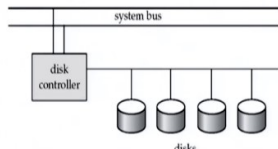
So, A is a track B is a geometrical sector, the which is shown in blue, C is a track sector as we are showing in orange and D is what we say is a cluster or block. A cluster or block is basically a set of consecutive sectors on the same track, which you typically use together, write together, read together and so on. And this is one of these disks, that is how it goes back and you can read them.

So, this physical organization just is very important, and you should try to have this in your mind. I mean, not need to look up this diagram you should be able to close your eyes and see this diagram because this is very, very required in terms of doing the analysis. So, the read write head positioning, you have understood, these are the tracks and per platter there may be 50,000 to 100,000 tracks like this, as you go radially. Tracks into sectors you read write finally, every sector that is a unit. So, your disk head swings into that position. And once it get the position the sector is below it, it can be read or written. And cylinders, we have explained what the cylinders are those are virtual disks that are getting formed by one position of the arm assembly.

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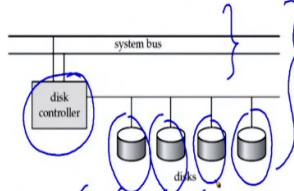
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Magnetic Disks (3): Disk Controller, Subsystems, and Interfaces

- **Disk Controller:** interfaces between the computer system and the disk drive hardware
 - Accepts high-level commands to read or write a sector
 - Initiates actions moving the disk arm to the right track, reading or writing the data
 - Computes and attaches checksums to each sector to verify that correct read back
 - Ensures successful writing by reading back sector after writing it
 - Performs remapping of bad sectors
- **Disk Subsystem:**

```
graph TD
    SB[system bus] --- DC[disk controller]
    DC --- D1[(disk)]
    DC --- D2[(disk)]
    DC --- D3[(disk)]
    DC --- D4[(disk)]
    subgraph disks
        D1
        D2
        D3
        D4
    end
```
- **Disk Interface Standards Families:** ATA, SATA, SCSI, SAS, several variants
- **Storage Area Networks (SAN)** connects disks by a high-speed network to a number of servers
- **Network Attached Storage (NAS)** provides a file system interface using networked file system protocol

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So, given this is the basic structure of a disk. What you typically have in the system is a disk controller. A disk controller is what controls this disk, it gets commands for reading or writing sectors, it initiates the movement of the arm to the right track, then it checks the rotation, so that you get to the right sector.

So, movement in rotation to get to the right sector and then you are able to read. And what it does, it to make sure that there are not errors you normally read write with a checksum. I mean, there is a mathematical, some mathematical formula used, so that from the entire data you compute a small number, so that if anything in the data changes and you compute it again this checksum will be different.

You could use MD5 hash or something of that sort. So, it ensures that success, I mean, every writing is successful, because first it writes and then it reads back, and it also writes the checksum, so in future when it is read, if after reading it computes the checksum and sees that the stored checksum is different, it knows that something has happened wrong, the sector has gone bad.

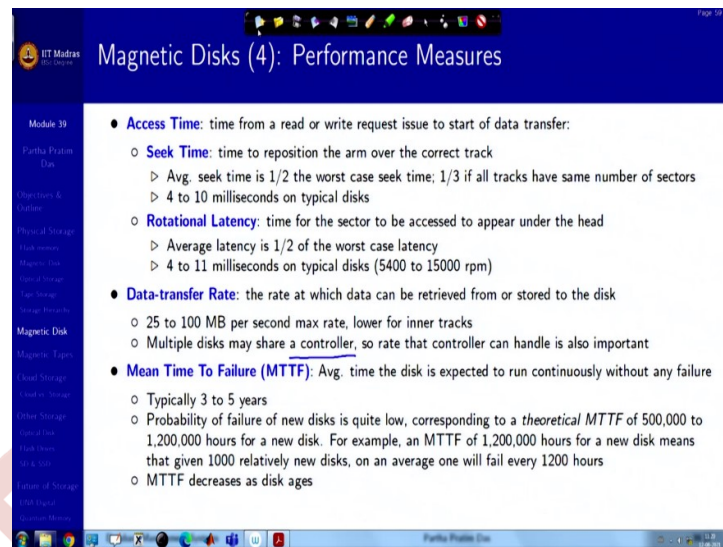
It might map, remap bad sectors because if some sectors go bad it will try to map them to others and so on. Other details I will not go into so much often. You will have a disk subsystem so that every disk may have its own controller for most of it for the spindles and all that and disk controller will operate at a higher level that way it becomes, that way it becomes much faster to have a disk assembly and then you have a system bus which is a very, very high-speed network connecting all these disks.

Naturally to be able to put any disk into the system you need a certain disk interface standards, so there are several standards that you have heard of. Getting into the disk interface standards is not something that we have time for, but just know that there is things like ATA SATA, SCSI and so on. And depending on how you connect them, you these days you have either SAN or NAS storage area networks or network attached storage.

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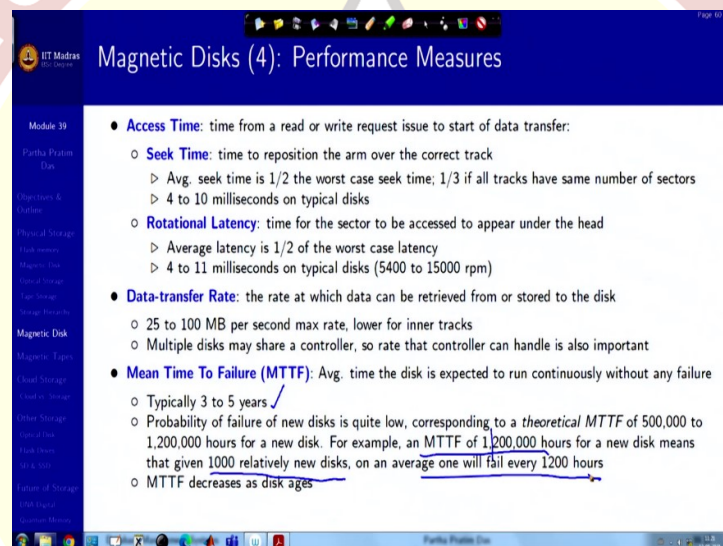
Magnetic Disks (4): Performance Measures

- **Access Time:** time from a read or write request issue to start of data transfer:
 - **Seek Time:** time to reposition the arm over the correct track
 - ▷ Avg. seek time is 1/2 the worst case seek time; 1/3 if all tracks have same number of sectors
 - ▷ 4 to 10 milliseconds on typical disks
 - **Rotational Latency:** time for the sector to be accessed to appear under the head
 - ▷ Average latency is 1/2 of the worst case latency
 - ▷ 4 to 11 milliseconds on typical disks (5400 to 15000 rpm)
- **Data-transfer Rate:** the rate at which data can be retrieved from or stored to the disk
 - 25 to 100 MB per second max rate, lower for inner tracks
 - Multiple disks may share a controller, so rate that controller can handle is also important
- **Mean Time To Failure (MTTF):** Avg. time the disk is expected to run continuously without any failure
 - Typically 3 to 5 years
 - Probability of failure of new disks is quite low, corresponding to a *theoretical MTTF* of 500,000 to 1,200,000 hours for a new disk. For example, an MTTF of 1,200,000 hours for a new disk means that given 1000 relatively new disks, on an average one will fail every 1200 hours
 - MTTF decreases as disk ages



Magnetic Disks (4): Performance Measures

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 - **Seek Time:** time to reposition the arm over the correct track
 - ▷ Avg. seek time is 1/2 the worst case seek time; 1/3 if all tracks have same number of sectors
 - ▷ 4 to 10 milliseconds on typical disks
 - **Rotational Latency:** time for the sector to be accessed to appear under the head
 - ▷ Average latency is 1/2 of the worst case latency
 - ▷ 4 to 11 milliseconds on typical disks (5400 to 15000 rpm)
- **Data-transfer Rate:** the rate at which data can be retrieved from or stored to the disk
 - 25 to 100 MB per second max rate, lower for inner tracks
 - Multiple disks may share a controller, so rate that controller can handle is also important
- **Mean Time To Failure (MTTF):** Avg. time the disk is expected to run continuously without any failure
 - Typically 3 to 5 years
 - Probability of failure of new disks is quite low, corresponding to a *theoretical MTTF* of 500,000 to 1,200,000 hours for a new disk. For example, an MTTF of 1,200,000 hours for a new disk means that given 1000 relatively new disks, on an average one will fail every 1200 hours
 - MTTF decreases as disk ages

So, this is the overall structure of a magnetic disk or the disk system that we typically have for database. Another question is, what are the, I mean, what are the metrics for performance? Obviously, the first metric is access time, the time to read or write a data, which is the time from the request, the issue of the request to the start of the data transfer. Why start of the data transfer? Because the remaining time does not depend on the disk, it depends on the channel you have between the disk and the actual main memory, which is a system bus and other systems.

Now, this access time is again divided into two parts. One is a seek time, because just recall there is a set of disk head and the disk is rotating. So, first it has to align for the track. So, first it has to align for the track, it has to decide in which track it is there, so it has to seek for the track.

Once it has got the track then on rotation it has to get to the write sector, the tracks are this way, sectors or this way. So, seek time is to find the track, rotational latency is to within a track get to that particular sector. So, typically, the average seek time happens to be half of the worst case seek time and it is 4 to 11 milliseconds on typical disks and the rotation latency is also of the same order assuming that what RPM rotations per minute you are spinning the disk.

So, this is a primary effect then is that net throughput metric the data transfer rate. Having given all that, if I have a chunk of data to transfer, then what is the rate at which I can transfer the data, so that will be typically 25, to 100 megabytes per second maximum rate. Naturally, lower tracks have less data, so there the overhead of seek time and rotational latency is more than the outer tracks.

So, inner tracks will have less transfer rate than the outer tracks, and multiple disks may share a controller I showed you the disk system configuration. So, the rate is also dependent on how what strategy the controller is taking and how it manages it. The third and so, we talked about raw axis time, the transfer rate and then the reliability question is known as MTTF Mean Time To Failure. That is, from one failure to the next how much time would elapse So, how often will this disk system fail or in other words, what is the average time for which it can run without failure continuously?

So, typically for disks today it is 3 to 5 years. And the probability of new disks are failing is very, very low, which is kind of it is expressed in terms of an almost million of hours. So, if we say that an MTTF of a new disk is 1, 2, 3, 1.2 million, 1 to 2, 1.2 million hours, then it means that the 1000, relatively new disks on the average will fail one of those. If I have 1000, relatively new disks, and one of those will fail every 1,200 hours. So, naturally it is needless to mention that MTTF will decrease as the disk ages. So, these are the three metrics to keep in mind, we might have to do some computations with them as well.

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Magnetic Tapes

- Hold large volumes of data and provide high transfer rates
 - Tape Formats
 - ▷ Few GB for DAT (Digital Audio Tape) format
 - ▷ 10-40 GB with DLT (Digital Linear Tape) format
 - ▷ 100 GB+ with Ultrium format, and
 - ▷ 330 GB with Ampex helical scan format
 - Transfer rates from few to 10's of MB/s
- Tapes are cheap, but cost of drives is very high
- Very slow access time in comparison to magnetic and optical disks
 - Limited to sequential access ✓
 - Some formats (Accellis) provide faster seek (10's of seconds) at cost of lower capacity
- Used mainly for backup, for storage of infrequently used information, and as an off-line medium for transferring information from one system to another.
- Tape jukeboxes used for very large capacity storage
 - Multiple petabytes (10^{15} bytes)

Now, the next important component is in the tertiary sector, which you have magnetic tapes, as I said, they are linear they are tapes. And therefore, they can hold large volume of data, provide high transfer rates, but nothing compared to what the disks can, but they are cheap compared to the disk, so you can keep really large amount of them. There are different formats available that you may have heard of digital audio tape, which are also used for audio recording. And then you have higher and higher formats, which can have higher and higher orders of data storage.

The sequentiality is predominant, and therefore, your access is slow, but it is good as a tertiary storage. And some of the recent disks are getting into couple of multiple of petabytes

I mean tape storage are getting into couple of petabytes as well. So, as a tertiary for a permanency of storage of historical data, magnetic tapes are really good.

(Refer Slide Time: 23:17)

The image displays two sequential screenshots of a presentation slide titled "Cloud Storage" from IIT Madras. The top screenshot shows the title slide with the text "Cloud Storage" in red. The bottom screenshot shows a list of bullet points about cloud storage, with a small video inset of a man in the bottom right corner.

Cloud Storage

- Cloud storage is purchased from a third-party cloud vendor who owns and operates data storage capacity and delivers it over the Internet in a pay-as-you-go model
- These cloud storage vendors manage capacity, security and durability to make data accessible to applications all around the world
- Applications access cloud storage through traditional storage protocols or directly via an API
- Many vendors offer complementary services designed to help collect, manage, secure and analyze data at massive scale. Various available options for cloud storage are:
 - Google Drive
 - Amazon Drive
 - OneDrive by Microsoft
 - Evernote
 - Dropbox
 - and so on

Cloud Storage

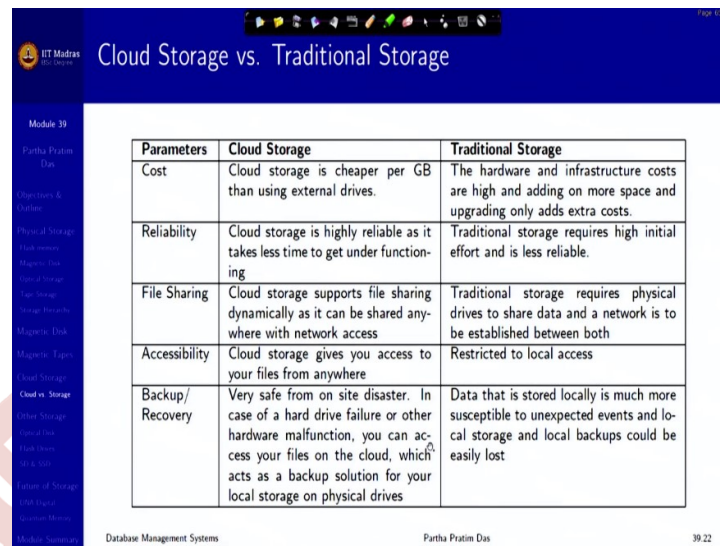
- Cloud storage is purchased from a third-party cloud vendor who owns and operates data storage capacity and delivers it over the Internet in a pay-as-you-go model
- These cloud storage vendors manage capacity, security and durability to make data accessible to applications all around the world
- Applications access cloud storage through traditional storage protocols or directly via an API
- Many vendors offer complementary services designed to help collect, manage, secure and analyze data at massive scale. Various available options for cloud storage are:
 - Google Drive ✓
 - Amazon Drive ✓
 - OneDrive by Microsoft ✓
 - Evernote ✓
 - Dropbox ✓
 - and so on

So, that was the core of what your database physical storage talk about. I will quickly take you through some of the other, I mean, available options because not always you are making an enterprise solution. So, if you are making a kind of a limited solution or if you can really make arrangements now a good option that has come up, which is more I should say it is technologically not very different, but it is significantly different business wise because it gives you a lot of options.

This is different forms of cloud storage. I mean, I am sure all of you are now familiar with it, we just uploading and spreadsheet and the Google Drive and saying, everybody can use it and Google gives that free up to a certain point. So that is a cloud storage. So, cloud storage is getting very, very popular in terms of the physical storage. And often times up to a certain extent, you have complimentary services. So, if you are making a personal database application or small enterprise database application then you may be able to even manage without paying and have your databases there.

Obviously, they will never be as efficient as having your data onsite, but for that, there are other options as well. For example, you could also run your entire data application on that same cloud, so that it is remotely, but the main point is, it will collect, manage, secure analyze data at massive scale for you. So, some of the options that you have is obviously Google Drive, Amazon drive, OneDrive, Evernote, Dropbox and so on, they give you varied options of cloud storage, which is really useful.

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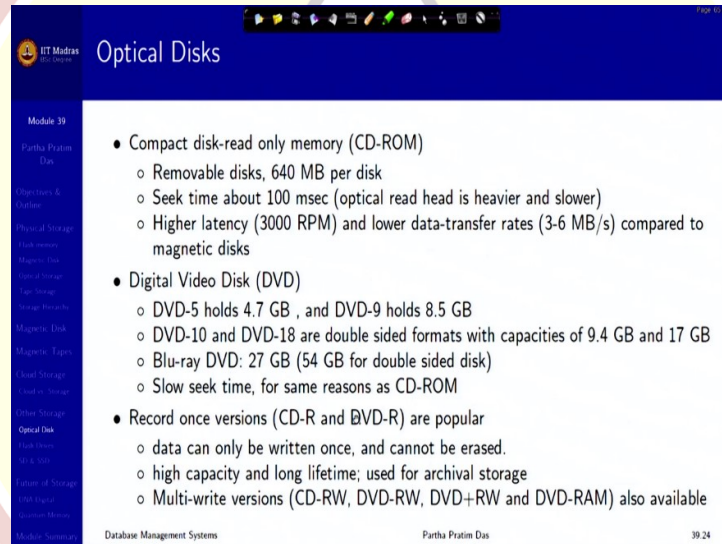
Parameters	Cloud Storage	Traditional Storage
Cost	Cloud storage is cheaper per GB than using external drives.	The hardware and infrastructure costs are high and adding on more space and upgrading only adds extra costs.
Reliability	Cloud storage is highly reliable as it takes less time to get under functioning	Traditional storage requires high initial effort and is less reliable.
File Sharing	Cloud storage supports file sharing dynamically as it can be shared anywhere with network access	Traditional storage requires physical drives to share data and a network is to be established between both
Accessibility	Cloud storage gives you access to your files from anywhere	Restricted to local access
Backup/ Recovery	Very safe from on site disaster. In case of a hard drive failure or other hardware malfunction, you can access your files on the cloud, which acts as a backup solution for your local storage on physical drives	Data that is stored locally is much more susceptible to unexpected events and local storage and local backups could be easily lost

So, here is just a, point wise comparison, cloud storage is usually less expensive. Though it might look like that you are having to pay every month, but if you look at the upfront investment you need to do for a traditional storage and the overhead of maintenance and stuff and all that cloud storage often is cost effective. It is reliable, because there is a lot of high-end disk technologies like RAID, and all those are used to make sure that it does not fail. File sharing is good accessibility is from anywhere, so that gives you a lot of better functionality automatically to your, database application. And backup recovery is completely handled by the cloud storage provider and it is safe from a site disaster. So, lot of advantages. So more and more, I am sure the database applications are migrating towards cloud and that is a way to go.

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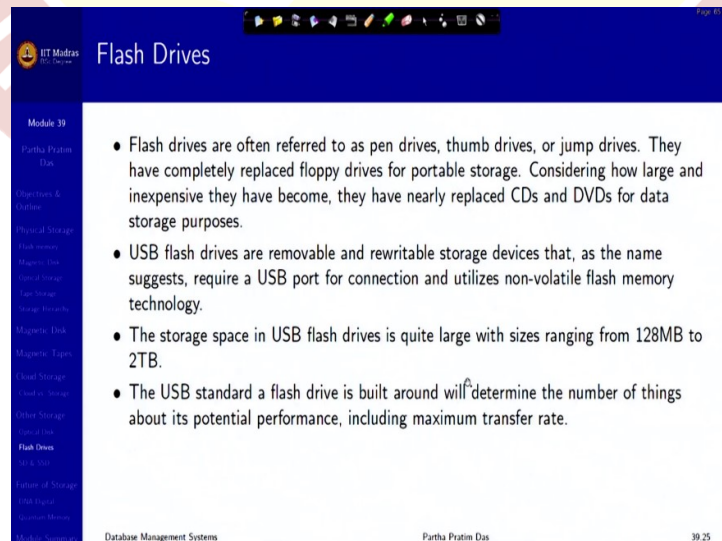
The slide is titled "Other Storage" in a large, bold, red font. It is part of a presentation from IIT Madras, as indicated by the logo in the top left corner. The slide number "39.23" is in the bottom right corner. The left sidebar contains a list of topics: Module 39, Partha Pratim Das, Objectives & Outline, Physical Storage, Disk memory, Magnetic Disk, Optical Storage, Tape Storage, Storage Protocols, Magnetic Disk, Magnetic Tape, Cloud Storage, and Other Storage (which is highlighted). The bottom of the slide shows "Database Management Systems" and "Partha Pratim Das".



The slide is titled "Optical Disks" in a large, bold, white font. It is part of a presentation from IIT Madras, as indicated by the logo in the top left corner. The slide number "39.24" is in the bottom right corner. The left sidebar contains a list of topics: Module 39, Partha Pratim Das, Objectives & Outline, Physical Storage, Disk memory, Magnetic Disk, Optical Storage (which is highlighted), Tape Storage, Storage Protocols, Magnetic Disk, Magnetic Tape, Cloud Storage, and Other Storage. The main content area lists the following:

- Compact disk-read only memory (CD-ROM)
 - Removable disks, 640 MB per disk
 - Seek time about 100 msec (optical read head is heavier and slower)
 - Higher latency (3000 RPM) and lower data-transfer rates (3-6 MB/s) compared to magnetic disks
- Digital Video Disk (DVD)
 - DVD-5 holds 4.7 GB, and DVD-9 holds 8.5 GB
 - DVD-10 and DVD-18 are double sided formats with capacities of 9.4 GB and 17 GB
 - Blu-ray DVD: 27 GB (54 GB for double sided disk)
 - Slow seek time, for same reasons as CD-ROM
- Record once versions (CD-R and DVD-R) are popular
 - data can only be written once, and cannot be erased.
 - high capacity and long lifetime; used for archival storage
 - Multi-write versions (CD-RW, DVD-RW, DVD+RW and DVD-RAM) also available

The bottom of the slide shows "Database Management Systems" and "Partha Pratim Das".



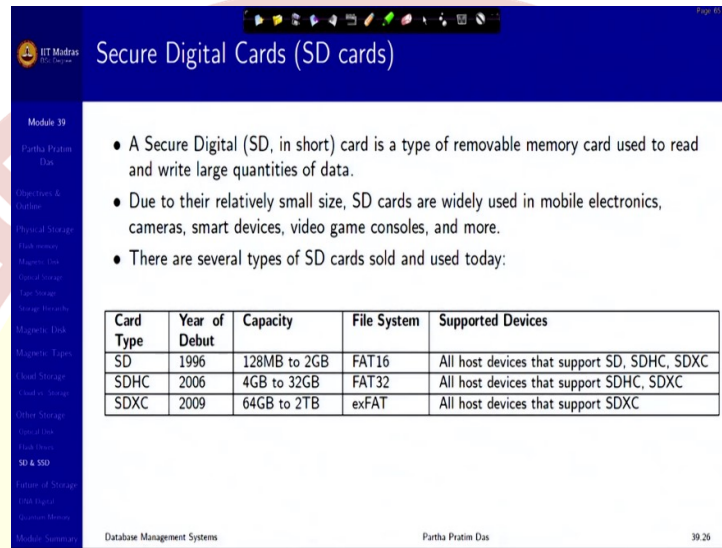
The slide is titled "Flash Drives" in a large, bold, white font. It is part of a presentation from IIT Madras, as indicated by the logo in the top left corner. The slide number "39.25" is in the bottom right corner. The left sidebar contains a list of topics: Module 39, Partha Pratim Das, Objectives & Outline, Physical Storage, Disk memory, Magnetic Disk, Optical Storage, Tape Storage, Storage Protocols, Magnetic Disk, Magnetic Tape, Cloud Storage, and Flash Drives (which is highlighted). The main content area lists the following:

- Flash drives are often referred to as pen drives, thumb drives, or jump drives. They have completely replaced floppy drives for portable storage. Considering how large and inexpensive they have become, they have nearly replaced CDs and DVDs for data storage purposes.
- USB flash drives are removable and rewritable storage devices that, as the name suggests, require a USB port for connection and utilizes non-volatile flash memory technology.
- The storage space in USB flash drives is quite large with sizes ranging from 128MB to 2TB.
- The USB standard a flash drive is built around will determine the number of things about its potential performance, including maximum transfer rate.

The bottom of the slide shows "Database Management Systems" and "Partha Pratim Das".

There are several other storages like optical disk we talked about, which are primarily for tertiary or for other applications. There are flash drives, which are we all are often using. So, they are good in terms of the fact that they are electronic, permanent storage. There is no moving part, there is no magnetism, it is just that electronically stored. They are not volatile, they can remember what you write, so those are advantages in small scale application.

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Secure Digital Cards (SD cards)

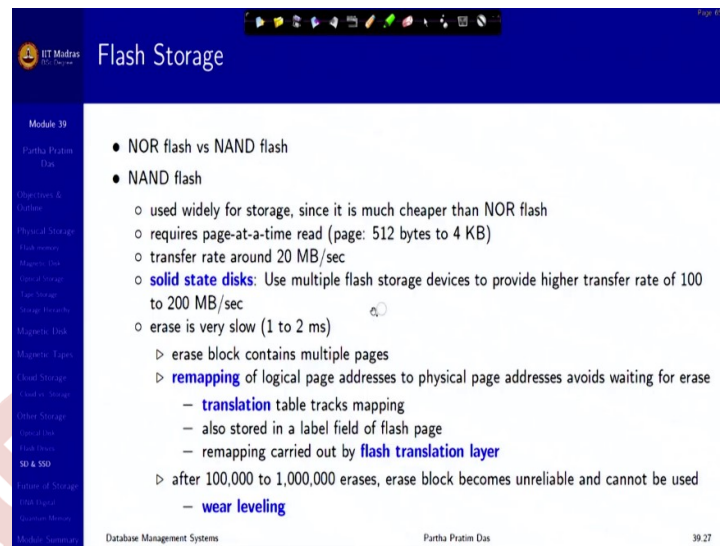
- A Secure Digital (SD, in short) card is a type of removable memory card used to read and write large quantities of data.
- Due to their relatively small size, SD cards are widely used in mobile electronics, cameras, smart devices, video game consoles, and more.
- There are several types of SD cards sold and used today:

Card Type	Year of Debut	Capacity	File System	Supported Devices
SD	1996	128MB to 2GB	FAT16	All host devices that support SD, SDHC, SDXC
SDHC	2006	4GB to 32GB	FAT32	All host devices that support SDHC, SDXC
SDXC	2009	64GB to 2TB	exFAT	All host devices that support SDXC

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Others, which you are very familiar with, I am sure you are regularly pushing in and out to your mobile phone or SD cards, which are called Secure Digital cards, which are also, storage of certain small size, but they are getting popular in terms of consideration for a database design application, because you could have a database, small database application running on your handheld device where your secondary storage is primarily the SD cards, that is very, very possible.

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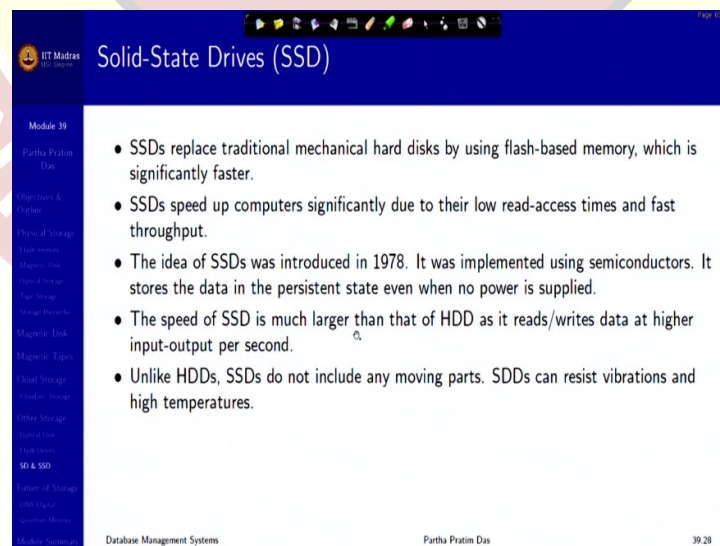
Flash Storage

- NOR flash vs NAND flash
- NAND flash
 - used widely for storage, since it is much cheaper than NOR flash
 - requires page-at-a-time read (page: 512 bytes to 4 KB)
 - transfer rate around 20 MB/sec
 - **solid state disks**: Use multiple flash storage devices to provide higher transfer rate of 100 to 200 MB/sec
 - erase is very slow (1 to 2 ms)
 - ▷ erase block contains multiple pages
 - ▷ **remapping** of logical page addresses to physical page addresses avoids waiting for erase
 - **translation** table tracks mapping
 - also stored in a label field of flash page
 - remapping carried out by **flash translation layer**
 - ▷ after 100,000 to 1,000,000 erases, erase block becomes unreliable and cannot be used
 - **wear leveling**

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They are typically on different flash storage technologies. I will not go into those, those are deeper in electronics, but these are some of the features that you have and there are NOR flash technology and NAND flash technology of which NAND is more popular. I should say is more reliable and so on and it has a lot of advantages, but it is a flash, so it you can it is conceptually write-only so you can write then erase and then write it again, but it does go bad after. So, you see that your pendrive is not working after getting old for a year or two, that is basically often the reason is you have run out of your erase cycles.

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Solid-State Drives (SSD)

- SSDs replace traditional mechanical hard disks by using flash-based memory, which is significantly faster.
- SSDs speed up computers significantly due to their low read-access times and fast throughput.
- The idea of SSDs was introduced in 1978. It was implemented using semiconductors. It stores the data in the persistent state even when no power is supplied.
- The speed of SSD is much larger than that of HDD as it reads/writes data at higher input-output per second.
- Unlike HDDs, SSDs do not include any moving parts. SSDs can resist vibrations and high temperatures.

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There is a in this class, there exists solid state drives, which are basically flash-based memory not the disk one. It is little it is a little bigger than the usual flash memory we use. But the advantage is it is very, very fast. It is very, very fast and it is much larger than the main memory so can use it for the purpose of a disk.

So, instead of HDD, you can have SSD Solid State Device and we are often getting devices like laptops and so on with the SSD having your BIOS and the basic operating system. So, which makes that it turns on the laptop, which used to turn on with HDD with maybe 3 minutes will now turn on in 30 seconds or even below that, but it is certainly quite expensive. And there are some comparative parameters of a SSD and HDD I have presented here this is more for your reference.

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Future of Storage

Page No. _____

Module 39

Partha Pratim Das

Objectives & Outcomes

Physical Storage

Flash memory

Magnetic Disk

State of Storage

File Storage

Storage Protocols

Magnetic Disk

Magnetic Tapes

Cloud Storage

Other Storage

Future of Storage

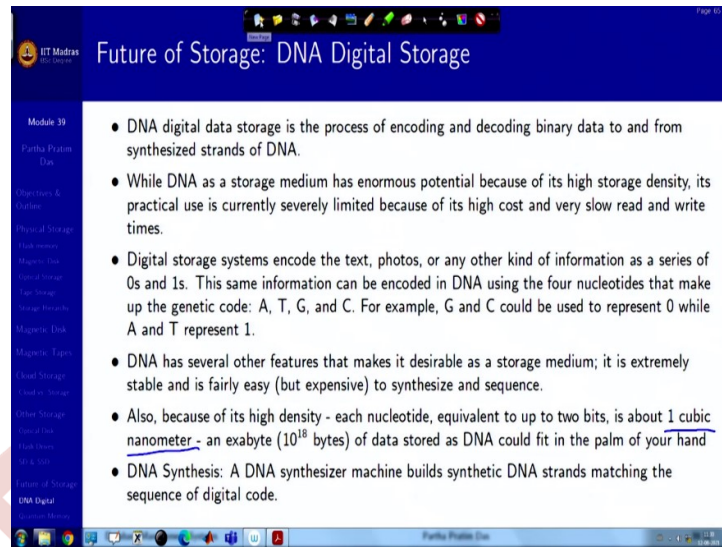
Future of Storage

Future of Storage

Future of Storage

Partha Pratim Das

Database Management Systems



Future of Storage: DNA Digital Storage

- DNA digital data storage is the process of encoding and decoding binary data to and from synthesized strands of DNA.
- While DNA as a storage medium has enormous potential because of its high storage density, its practical use is currently severely limited because of its high cost and very slow read and write times.
- Digital storage systems encode the text, photos, or any other kind of information as a series of 0s and 1s. This same information can be encoded in DNA using the four nucleotides that make up the genetic code: A, T, G, and C. For example, G and C could be used to represent 0 while A and T represent 1.
- DNA has several other features that makes it desirable as a storage medium; it is extremely stable and is fairly easy (but expensive) to synthesize and sequence.
- Also, because of its high density - each nucleotide, equivalent to up to two bits, is about 1 cubic nanometer - an exabyte (10^{18} bytes) of data stored as DNA could fit in the palm of your hand
- DNA Synthesis: A DNA synthesizer machine builds synthetic DNA strands matching the sequence of digital code.

Well, the developments keep on happening. There will be lot of advances in magnetic optical and electronic memory, so that is a storage that is going to be there. But two major you know trends that are emerging very strongly is one, which is called DNA digital storage. Our DNA has a lot of memory, it stores a lot of information. So, can synthesize strands of DNA be used as storage. So, DNA have a A, T, G, C coding can similar kind of coding in a synthesize DNA. So, this is a biological kind of memory. So, you will have nucleotides and in one cubic-nanometer.

So, one cubic nanometer you can think and exabyte of data can be I mean a very high density of data can be stored. So, that an exabyte of data can be stored as DNA which will come on the palm of your hand, which is other, not possible by other technologies. It is still very expensive, it is still slow. There is not a commercial stage, but it has a lot of potential.

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Future of Storage: Quantum Memory

- Quantum memory is the quantum-mechanical version of ordinary computer memory
- Whereas ordinary memory stores information as binary states (represented by "1"s and "0"s), quantum memory stores a quantum state for later retrieval
- These states hold useful computational information known as *qubits*
- Quantum memory is essential for the development of many devices in quantum information processing applications such as quantum network, quantum repeater, linear optical quantum computation or long-distance quantum communication
- Unlike the classical memory of everyday computers, the states stored in quantum memory can be in a quantum superposition, giving much more practical flexibility in quantum algorithms than classical information storage

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Last, but not the least is the advances in the quantum technology. So, you know that quantum mechanical versions of computer memory is coming up quantum version of computing is coming up where bits are not there, there are rather Q bits the quantum bits and this memory is also going to is expected to come up to a good size and efficiency in the next couple of years.

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Module Summary

- Understood the range of Physical Storage Media
- Studied the mechanism and performance of the Magnetic Disks
- Looked at the features of Magnetic Tape as tertiary storage
- Glimpsed through Other Storage including Optical Disk, Flash and SSD
- Considered the Future of Storage in terms of DNA and Quantum

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Edited and new slides are marked with "PPD".

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So, we have taken a entire look at the physical storage media. We will move on to getting more focused in terms of our database, backend I mean, database back end physical organization part of it. Thank you very much for your attention, and meet you in the next module.